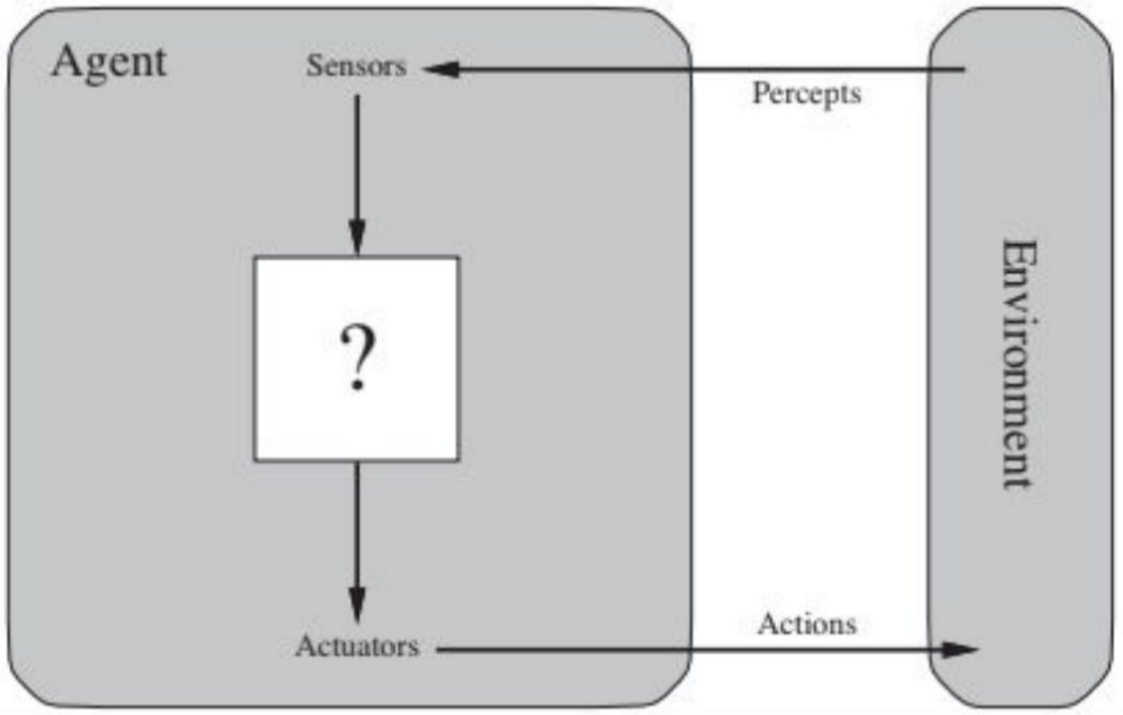
Agents

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators



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Agents

- An Agent runs in the cycle of perceiving, thinking, and acting. An agent can be:
  - Human-Agent: A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
  - Robotic Agent: A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
  - Software Agent: Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.
- The term percept refer to the agent's perceptual inputs at any given instant.
- An agent's percept sequence is the complete history of everything the agent has ever perceived.
- In general, an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived.

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Agent Functions

- An agent's behavior is described by the agent function .
- The agent function maps from percept histories to actions.
- The agent function for an artificial agent will be implemented by an program.
- The agent program runs on the physical architecture to produce the agent function.
- The agent function is an abstract mathematical description; the agent program is a concrete implementation, running within some physical system.
- agent = architecture + program

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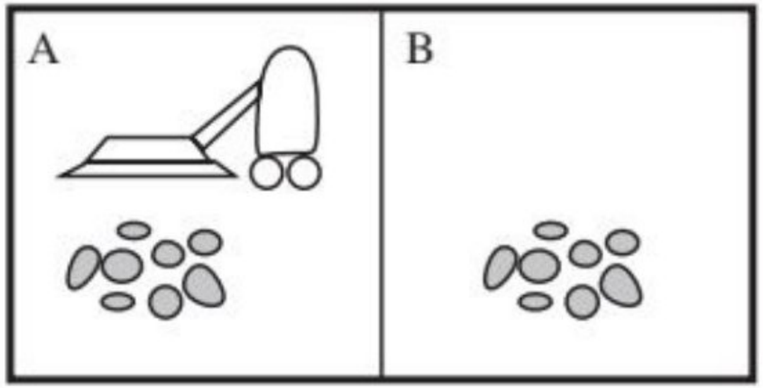
The Vacuum Cleaner World

- This particular world has just two locations: squares A and B.
- The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing.
- One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.
- A partial tabulation of this agent function is shown in Figure.



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Percept sequence	Action
$[A, Clean]$	*Right*
$[A, Dirty]$	*Suck*
$[B, Clean]$	*Left*
$[B, Dirty]$	*Suck*
$[A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Dirty]$	*Suck*
⋮	⋮
$[A, Clean], [A, Clean], [A, Clean]$	*Right*
$[A, Clean], [A, Clean], [A, Dirty]$	*Suck*
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

- And an agent program that implements it appears in Figure
- Percepts: location and contents, e.g., [A,Dirty]
- Actions: Left, Right, Suck, NoOp
- Agent's function → look-up table

```
function REFLEX-VACUUM-AGENT([location,status]) returns an action
  if status = Dirty then return Suck
  else if location = A then return Right
  else if location = B then return Left
```

Figure 2.8 The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 2.3.

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Intelligent Agents

An intelligent agent is an autonomous entity which act upon an environment using sensors and actuators for achieving goals.

An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI Agent

- Rule 1: An AI agent must have the ability to perceive the environment.
- Rule 2: The observation must be used to make decisions.
- Rule 3: Decision should result in an action.
- Rule 4: The action taken by an AI agent must be a rational action.

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Good behavior: The concept of rationality

- A **rational agent** is said to perform the right things. AI is about creating rational agents to use for game theory and decision theory for various real-world scenarios.
- A rational agent is an agent which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions.
- This notion of desirability is captured by a performance measure that evaluates any given sequence of environment states.

- For the vacuum cleaner problem , We might propose to measure performance by the amount of dirt cleaned up in a single eight hour shift.
- A rational agent can maximize this performance measure by cleaning up the dirt, then dumping it all on the floor, then cleaning it up again, and so on.
- A more suitable performance measure would reward the agent for having a clean floor. For example, one point could be awarded for each clean square at each time step (perhaps with a penalty for electricity consumed and noise generated).

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The nature of Environments

- Rationality can be judged on the basis PEAS System.
- Task environments, are essentially the “problems” to which rational agents are the “solutions.”
- We can call it as PEAS which stands for a Performance measure, Environment, Actuator, Sensor.
- PEAS System is used to categorize similar agents together. Most of the highest performing agents are Rational Agents.
- PEAS for self-driving cars:
    - Performance: Safety, time, legal drive, comfort
    - Environment: Roads, other vehicles, road signs, pedestrian
    - Actuators: Steering, accelerator, brake, signal, horn
    - Sensors: Camera, GPS, speedometer, odometer, accelerometer, sonar.

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Examples of agent types & PEAS description

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.

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Properties of task environments

An environment in artificial intelligence is the surrounding of the agent. The agent takes input from the environment through sensors and delivers the output to the environment through actuators. There are several types of environments:

- Fully Observable vs Partially Observable
- Deterministic vs Stochastic
- Single-agent vs Multi-agent
- Static vs Dynamic
- Discrete vs Continuous
- Episodic vs Sequential
- Known vs Unknown

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Fully Observable vs Partially Observable

- When an agent sensor is capable to sense or access the complete state of an agent at each point in time, it is said to be a fully observable environment else it is partially observable.
- Maintaining a fully observable environment is easy as there is no need to keep track of the history of the surrounding.
- An environment is called unobservable when the agent has no sensors in all Environments.
    - Chess – the board is fully observable, and so are the opponent's moves.
    - Driving – this partially observable since what's around the corner is unknown

Deterministic vs Stochastic

- When a uniqueness in the agent's current state completely determines the next state of the agent, the environment is said to be deterministic.
- The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.
    - Chess – there would be only a few possible moves for a coin at the current state and these moves can be determined.
    - Self-Driving Cars- the actions of a self-driving car are not unique, it varies time to time.

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Single-agent vs Multi-agent

- An environment consisting of only one agent is said to be a single-agent environment.
- A person left alone in a maze is an example of the single-agent system.
- An environment involving more than one agent is a multi-agent environment.
- The game of football is multi-agent as it involves 11 players in each team.
    - Chess – chess is in a two- agent environment.
    - Driving – multi agent as all other cars must be treated as agents

Dynamic vs Static

- An environment that keeps constantly changing itself when the agent is up with some
- action is said to be dynamic.
- A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.
- An idle environment with no change in its state is called a static environment.
- An empty house is static as there's no change in the surroundings when an agent enters.
    - Taxi driving is clearly dynamic: the other cars and the taxi itself keep moving while the driving algorithm dithers about what to do next.
    - Crossword puzzles are static.

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Discrete vs Continuous

- If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.
- The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.
- Chess : is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it's finite.
- Self-driving cars : continuous environments as their actions are driving, parking, etc. which cannot be numbered.

Known vs Unknown

- Here, "known" or "unknown" don't describe the environment, but instead describe the agent's perception of the environment.
- In a **known** environment, an agent is aware of the available actions and their consequences in terms of pursuing the objective(s). This can also be considered understanding the rules of the objective it is pursuing. For many adults, tic-tac-toe is a known environment. They know what they need to do to win and how their actions will affect those winnings on a probability basis.

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- In an **unknown** environment, an agent does not know what reactions the actions it performs will cause. Think of a human entering the world. They must be taught, or figure out through experience, every facet of life. This is also an example of how the environment may be unknown to one agent but known to another (a professional piano player playing a song).

Episodic vs Sequential

- In an Episodic task environment, each of the agent's actions is divided into atomic incidents or episodes. There is no dependency between current and previous incidents. In each incident, an agent receives input from the environment and then performs the corresponding action.
- Example: Consider an example of Pick and Place robot, which is used to detect defective parts from the conveyor belts. Here, every time robot (agent) will make the decision on the current part i.e. there is no dependency between current and previous decisions.
- In a Sequential environment, the previous decisions can affect all future decisions. The next action of the agent depends on what action he has taken previously and what action he is supposed to take in the future.
    - Checkers- Where the previous move can affect all the following moves.

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Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

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